



UNIVERSITY OF THE PUNJAB

Roll No.

Fifth Semester 2018
Examination: B.S. 4 Years Programme

PAPER: Probability Theory (Theory)
Course Code: STAT-307

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

OBJECTIVE TYPE

Q.1 Tick the correct answer for the following multiple choice questions.

- I. If X is number of trials for negative binomial distribution with parameters p and k then its minimum value is :
 - a. 0
 - b. k
 - c. $k+1$
 - d. $k-1$
- II. For a given binomial distribution with n fixed if $p=0.5$ then?
 - a. Poisson distribution will provide a bad approximation
 - b. Poisson distribution will provide a good approximation
 - c. Binomial distribution will be skewed left
 - d. Binomial distribution will be skewed right
- III. A fair coin is tossed 100 times, the expected number of heads are :
 - a. 100
 - b. 50
 - c. 15
 - d. None of these
- IV. Which of the following is a characteristic of the probability distribution for any random variable?
 - a. A probability is provided for every possible value
 - b. The sum of all probabilities is one
 - c. No given probability occurs more than once
 - d. Both (a) and (b)
- V. A binomial distribution may be approximated by a Poisson distribution if:
 - a. n is small and p is small
 - b. n is large and p is large
 - c. n is small and p is large
 - d. none of these
- VI. If a letter is chosen at random from the 10 letters of the word STATISTICS, what is probability that it is a vowel?
 - a. 0.20
 - b. 0.23
 - c. 0.30
 - d. 0.40

P.T.O.

VII. A and B are two independent events. The probability of A is $\frac{1}{4}$ and B is $\frac{1}{3}$. The neither probability of A nor B is

- a. $\frac{5}{12}$
- b. $\frac{1}{3}$
- c. $\frac{3}{4}$
- d. $\frac{11}{12}$

VIII. The number of ways to select to persons from 6, ignoring order of selection;

- a. 64
- b. 15
- c. 36

IX. A cricket team captain wins the toss for three consecutive matches. What is the probability that he will call correctly for the fourth march?

- a. $\frac{1}{16}$
- b. $\frac{1}{4}$
- c. $\frac{1}{8}$
- d. $\frac{1}{2}$

X. Which of the following statement is not true?

- a. Mutually exclusive events are statistically dependent
- b. Complementary events have probabilities that sum to one?
- c. Opposite events ate statistical independent.
- d. An experiment's elementary events are collectively exhaustive and mutually exclusive



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Fifth Semester 2018

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TIME ALLOWED: 2 hrs. & 30 mins.

MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Q.2 Write short answer on the following questions.

(20)

- What are the differences between a priori probability and empirical probability?
- If $P(A) = 0.7$, $P(B) = 0.6$, and events A and B are independent, find $P(A \text{ and } B)$
- What is the difference between mutually exclusive events and collectively exhaustive events?
- A coin is known to be unbalanced in such a way that heads only comes up 0.4 of the time.
 - What is the probability that first head appears on 4th toss?
 - How many toss would it take, on average, to flip two heads?
- What are the first four properties that must be present in order to use the negative binomial distribution?

Q.3.a) A sample of 500 respondents was selected in a large metropolitan area to study consumer behavior. Among the question asked was "Do you enjoy shopping for clothing?" Of 240 males, 136 answered yes. Of 260 females, 224 answered yes. What is the probability that a respondent chosen at random

- enjoys shopping for clothing?
- is a female and enjoys shopping for clothing?
- is a female or enjoy shopping for clothing?
- is a male or a female?

b) The probability that a person chosen has a certain disease is 0.03. Medical diagnostic tests are available to determine whether the person actually has the disease. If the disease is actually present, the probability that the medical diagnostic test will give a positive result is 0.90. If the disease is not actually present, the probability of a positive test result is 0.02. Suppose that the medical diagnostic test has given a positive result. What is the probability that the disease is actually present? What is the probability of a positive test result?

(10)

Q.4 Compute the mean, variance, coefficient of skewness and kurtosis of geometric distribution.

(10)

Q.5.a) Derive the Poisson distribution as the limiting form of the binomial distribution, stating clearly the assumptions you make.

b) Three people each toss a coin and the odd man pays for the coffee. If the coins all show heads or all show tails, they are tossed again. What is the probability that a decision is reached in five tosses or fewer?

(10)